

RISK ASSESSMENT

Project Title: *Water Rocket Project*
 Launch Day: *May 20th 2025*
 Location: *Falmer Campus*
 Assessors Name: *Pablo Fernandez-Figares*
 Date of Assessment: *May 13th 2025*

1. Hazard Identification and Risk Assessment

Hazard	Likelihood 1-5	Severity 1-5	Total Risk Rating (Likelihood x Severity) 1-25	Who might be at risk?	Measures that must be taken	Who must take this measures
Bottle exploding due to over- pressuring	2	3	6	Participants, specially the one pressuring the bottle.	Don't pressure the bottle over the limit of 10 bar.	The group participant in charge of pressuring the bottle.
Dangerous shapes in the rocket	2	5	10	All the participants and observers	Debur the sharp edges and tips.	3D and CAD team.
Rocket impacting on any person.	2	5	10	All the participants and observers	Keep all the observers and participant away from the launching area. Use a stabilizer to keep the launch angle steady and safe.	Two members of the group will be assigned with the role of keeping a safe area. The teacher will be probably controlling this too.
Electric Shock	1	4	4	Anyone in contact with the rocket.	Keep the micro bit isolated from the water and other hazardous components. Ensure to be dry before touching the rocket.	Participants in charge of the assembly and anyone who may touch the rocket.
Unpredictable weather conditions	2	4	8	All the participants and observers	Consult the weather report before the launch day and take	Teachers and group members

					actions according to the risk produced by the weather, cancelling the launch if the safety of the students and teachers is at risk.	
Hazardous materials on assembly (Super glue)	3	4	12	Group members in charge of the assembly	Make sure that all the participants are wearing gloves or safety equipment. Keep the windows opened. When not using the hot glue guns, keep them at a safe distance, so no one gets burnt.	Group members in charge of the assembly

2. Total Risk Rating Levels

- **Low Risk (1-4):** Acceptable risks if measures are taken.
- **Moderate Risk (5-9):** Acceptable, require controls in place and measures to reduce the risk.
- **High Risk (10-16):** Acceptable, require significant controls in place and measures to reduce the risk.
- **Unacceptable (17-25):** The launch cannot proceed.

3. Emergency Procedures

- There will be a First Aid Kit on site.
- Contact the teachers if you have any problem.
- Nearest Hospital: Royal Sussex County Hospital – Eastern Rd, Brighton and Hove, BN2 5BE